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M^{3}IT: A Large-Scale Dataset towards Multi-Modal Multilingual Instruction Tuning

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Abstract

Instruction tuning has significantly advanced large language models (LLMs) such as ChatGPT, enabling them to align with human instructions across diverse tasks. However, progress in open vision-language models (VLMs) has been limited due to the scarcity of high-quality instruction datasets. To tackle this challenge and promote research in the vision-language field, we introduce the Multi-Modal, Multilingual Instruction Tuning (M^{3}IT) dataset, designed to optimize VLM alignment with human instructions. Our M^{3}IT dataset comprises 40 carefully curated datasets, including 2.4 million instances and 400 manually written task instructions, reformatted into a vision-to-text structure. Key tasks are translated into 80 languages with an advanced translation system, ensuring broader accessibility. M^{3}IT surpasses previous datasets regarding task coverage, instruction number and instance scale. Moreover, we develop Ying-VLM, a VLM model trained on our M^{3}IT dataset, showcasing its potential to answer complex questions requiring world knowledge, generalize to unseen video tasks, and comprehend unseen instructions in Chinese. We have open-sourced the dataset to encourage further research.[1]

1 Introduction

There has been a continuously increasing trend to develop intelligent assistants that can follow human instructions [3, 36, 37]. In the natural language processing (NLP) field, instruction tuning [35, 53] is a success paradigm that leverages large-scale well-formatted instances to align large language models (LLMs) to human instructions. By finetuning on instances with specific task descriptions, LLMs learn to follow the instruction to perform various tasks, and demonstrate strong generalization ability on unseen tasks [29]. Expanding beyond NLP, a general-purpose intelligent agent must encompass various modalities, such as vision, prompting recent efforts to investigate instruction tuning in vision-language domains [63, 28, 7]. To develop powerful vision-language models (VLMs), it is essential to have a well-constructed dataset that encompasses diverse vision-language tasks and aligns with human instructions. However, the instructional data supporting existing VLMs is either not publicly available (e.g., GPT-4) or offers limited task and language coverage (e.g., only tasks in English are considered). This scarcity of comprehensive datasets has impeded the progress of open vision-language models, highlighting the importance of multi-modal instruction tuning and the need for high-quality datasets.

[1] Our dataset is available at https://huggingface.co/datasets/MMInstruction/M3IT

Preprint. Under review.

In this paper, we aim to advance instruction tuning research in the multi-modal domain by introducing an open dataset M^{3}IT, a **M**ulti-**M**odal **M**ultilingual **I**nstruction **T**uning dataset, as an essential step towards building a versatile general-purpose assistant. We build this dataset by converting existing datasets into a unified vision-to-text schema with four stages: (1) manual instruction writing, (2) dataset pre-processing, (3) careful quality check and (4) dataset translation for key tasks. Our dataset encompasses a wide range of tasks, including classic image-text tasks such as image classification, visual question answering, and image captioning. Video-related tasks, such as video question-answering, are also incorporated to ensure comprehensive coverage across multiple modalities. We further integrate Chinese vision-language datasets with corresponding Chinese instructions. The resulting dataset compiles 40 diverse tasks and 400 instructions. Finally, key vision-language tasks are translated into 80 languages with a strong translation system, to support multilingual studies.

To evaluate the effectiveness of the proposed dataset, we develop a vision-language model, Ying-VLM, by integrating a strong vision encoder, BLIP-2 [23] with a large language model, Ziya-13B [61], derived from LLaMA [49]. Building on the successful approach of incorporating visual tokens as textual prompts in LLMs [7, 63, 28], we employ a two-stage training process: (1) the initial stage aligns vision features with text embeddings through image captioning on LAION400M [41], and (2) the second stage enhances the model by conducting instruction tuning on selected tasks of our dataset. Experimental results reveal that Ying-VLM surpasses strong baseline models in knowledgeable VQA tasks and exhibits improved generalization performance to unseen video and cross-lingual tasks. Further analysis indicates that the improved performance corresponds to increased tasks for instruction tuning, while the diversity of instructions also affects outcomes.

This paper presents two key contributions: (1) We introduce the open-source, large-scale Multi-modal, multilingual Instruction Tuning (M^{3}IT) dataset, designed to enable the development of general-purpose multi-modal agents. (2) We develop Ying-VLM, a visual assistant that excels in knowledgeable VQA tasks, demonstrates strong generalization to unseen video QA and Chinese multi-modal tasks, and offers valuable insights for future research.

2 Related Work

Table 1: Summary of multi-modal instruction tuning datasets.

Dataset	# Tasks	Multi-Lingual	# of Instances	Avg. # of Manual Instructions / Task	Open-Sourced
MiniGPT4	N / A	✗	5K	N / A	✓
LLaVA	3	✗	1.15M	N / A	✓
MultiModalGPT	3	✗	6K	5	✗
MultiInstruct	26	✗	$\sim$ 235K	5	✗
InstructBLIP	28	✗	$\sim$ 1.6M	9.7	✗
M^{3}IT (Ours)	40	✓	2.4M	10	✓

Our work draws inspiration from recent language instruction tuning benchmarks [53, 35], which have been proven effective for improving language models to obtain cross-task generalization ability [29, 52]. In this paper, we focus on exploring the instruction tuning paradigm from LLMs to multi-modal agents. Unlike text-only tasks, vision-language tasks generally have more diverse formats, which poses new challenges toward vision-language instruction tuning benchmarks.

To develop a general-purpose vision-language model, it is crucial to create high-quality multi-modal instruction tuning datasets encompassing diverse tasks, languages, and instructions. Several studies have investigated multi-modal instruction tuning for VLMs. LLaVA [28] and MiniGPT-4 [63] generate visual content-related dialog by incorporating image caption data into GPT-4/ChatGPT models. MultiInstruct [56] reformats a series of visual classification tasks into an instruction-tuning format, while InstructBLIP [7] adapts 28 existing image-to-text tasks. However, these datasets do not provide an ideal multi-modal instruction tuning dataset due to their limited (1) coverage of various task types in multi-modal fields, (2) diversity and quality of instances, and (3) inclusion of multiple languages for wide linguistic diversity. In this paper, we construct an improved multi-modal instruction tuning dataset by expanding task coverage to 40 datasets, supplementing instances with 10 manually written task instructions, and including tasks in different languages. Table 1 compares the characteristics of existing multi-modal instruction tuning datasets and M^{3}IT.

3 M^{3}IT: A Multi-Modal Multilingual Instruction Tuning Dataset

In this section, we introduce our proposed M^{3}IT dataset by first elaborating the dataset coverage (§ 3.1), followed by the details of the annotation process(§ 3.2). Finally, we present the dataset format and provide the statistics of the crafted datasets instructions(§ 3.3).

3.1 Task Coverage

Our dataset compiles diverse tasks of classical vision-language tasks, including captioning, visual question answering (VQA), visual conditioned generation, reasoning and classification.

Captioning This task aims to produce descriptions of the given images according to different needs. We include MS COCO [27] (the Karpathy split) for generic image descriptions. TextCaps [44] requires models to capture the text presented in the image and generate captions accordingly. Image-Paragraph-Captioning [21] focuses on generating detailed descriptions for images.

Reasoning This task evaluates specific reasoning capabilities. We incorporate CLEVR [19] and NLVR [46] for spatial reasoning, Visual Commonsense Reasoning (VCR) [60] for commonsense reasoning, Visual MRC [47] for reading comprehensive over images, and Winoground [48] for fine-grained semantics reasoning over text descriptions and image contents.

Visual Question Answering (VQA) This is the most widely studied multi-modal task, which requires the model to answer a given question based on the image correctly. Tasks include VQA v2 [15], Shapes VQA [1], DocVQA [33], OCR-VQA [34], ST-VQA [2], Text-VQA [45], and GQA [18].

Knowledgeable Visual Question Answering Unlike traditional VQA tasks focusing on the question relevant to the content image, knowledgeable visual question answer (KVQA) requires the model to draw upon outside knowledge to answer questions. We incorporate two outside knowledge VQA datasets: OK-VQA [32] and A-OK-VQA [42], ScienceQA [31] which contains multi-modal science questions, and ViQuAE [22] focusing on knowledge facts of named entities in images.

Classification This task involves classifying an image based on a given set of candidate labels. ImageNet [40], Grounded Object Identification (COCO-GOI) [27], COCO-Text [50], Image Text Matching (COCO-ITM) [27], e-SNLI-VE [20], Multi-modal Fact Checking (Mocheg) [58], and IQA [9] are included. Due to language model input length constraints, we reduce the number of options in some datasets with extensive candidate labels, such as ImageNet.

Generation Visual conditional general requires models to understand the visual content and make a composition meeting the task demand. We have Visual Storytelling (VIST) [17], Visual Dialog (VisDial) [8], and multi-modal machine translation Multi30k [10] in this category.

Chinese and multilingual Vision-Language Tasks To examine the effect of instruction tuning on different languages, we incorporate several Chinese vision-language tasks including FM-IQA [11] for VQA, COCO-CN [25] and Flickr8k-CN [24] for captioning, Chinese Food Net [4] for classification, and MMChat [62] for generation.

Video-Language Tasks Beyond the static images, we are interested in whether instruction tuning can also be applied to video-text tasks. We include the classic MSR-VTT datasets [55] for video captioning, MSRVTT-QA [54], ActivityNet-QA [59], iVQA [57] and MSVD-QA [54] for video question answering, Something-Something [14] for video action classification.

As shown in Figure 1, our dataset makes a wide coverage of the current existing visual-language and video-language benchmarks, enabling different skill sets for the language models, from simple image captioning to complicated reasoning based on the image even beyond the visual content.

3.2 Annotation Process

To build high-quality multi-modal instruction datasets, we rewrite various datasets into a vision-to-text format. The annotation process includes four steps: (1) writing instructions for each task, (2) structuring images and texts into a unified schema, (3) checking the overall dataset quality, and (4) building multilingual sets. Eight authors of this work are employed as human annotators, each of whom is a graduate student familiar with relevant literature.

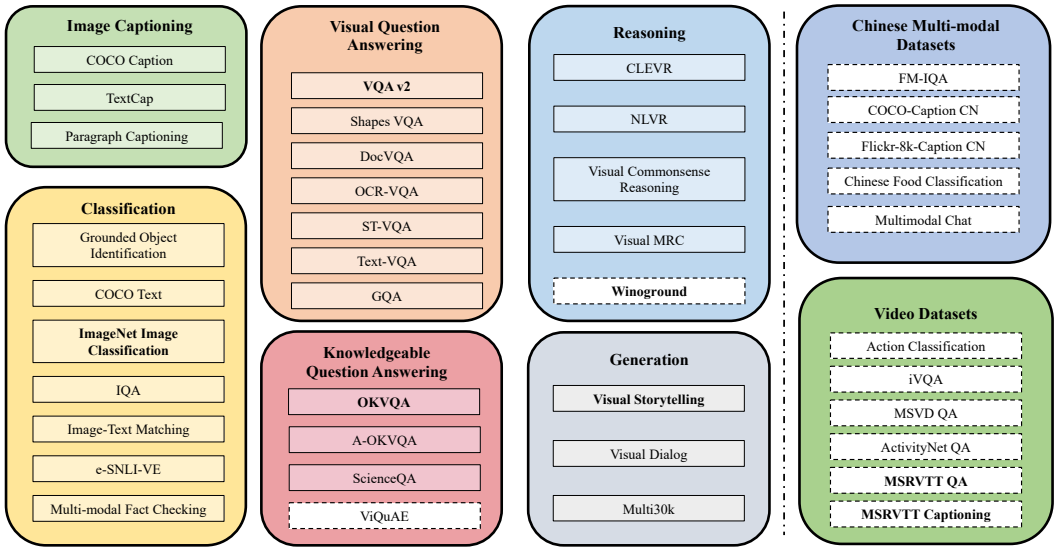

Figure 1: Tasks in our proposed multi-modal multilingual instruction tuning dataset. The tasks in dashed white boxes are held-out evaluation sets that are not adopted during training. Tasks with bold names are translated into 80 languages.

Table 2: The statistics of our instructions.

Number of different instructions	400
- Image Captioning	52
- Classification	113
- Visual Question Answering	95
- Knowledgeable Visual QA	40
- Reasoning	60
- Generation	40
Tokens per instruction	24.4 ± 9.6
Instruction edit distance among the same task	76.6 ± 37.2
Instruction edit distance across tasks	106.6 ± 39.5

Stage I: Instruction Writing To build high-quality instructions, we first ask annotators to carefully read the dataset paper and check the original dataset with some instances to get a clear understanding of the task. After that, they are required to write 10 diverse task instructions manually, covering the key characteristics of the task. Table 2 shows the statistics of the written instructions for each task. In total, we annotate 400 instructions for all tasks. The average length per instruction is 24.4. To evaluate the diversity of annotated instructions, we employ the average edit distance to measure the similarity between two strings. The average edit distance within the same task is 76.6, indicating a good range of instruction diversity.

Stage II: Data Format Unification After the instruction has been written according to the task characteristics, we further process the images and corresponding text for a unified instance schema. For most datasets, we keep the original images and text, where images are converted into corresponding base64 encoded strings for easy data loading. We perform two modifications on potential examples: (1) **Adding Bounding Box to Images.** For tasks designed for specific regions in the image, a straightforward solution is to provide the bounding box information in natural language for informing the language models of the regions in interest. However, the image preprocessing techniques adopted by different vision encoders may resize the original image, and the original bounding box annotation thus needs further adjustments. Inspired by the recent observation that common vision encoders such as CLIP [39] are sensitive to the visual prompt [43], we directly tag the bounding box as a red rectangle to the image, serving as a hint for VLMs to focus on the target region. (2) **Short Answer Paraphrasing.** As recent studies have shown that the original short and brief answers in the common

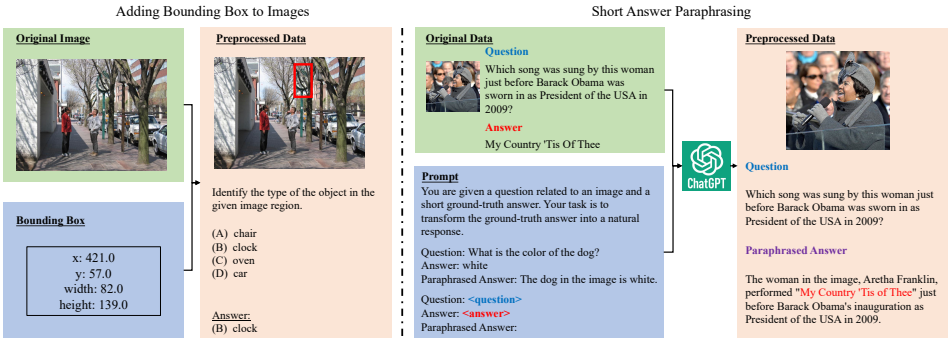

Figure 2: (Left) On region-based tasks, bounding boxes are added to original images to inform the model of the area in interest. (Right) Short answer paraphrasing to improve the response quality.

VQA dataset could negatively influence the model generation performance [7], we propose to utilize the ChatGPT [36] model for paraphrasing the original answers, by providing origin question and answer with potential extra contextual information. Contextual information includes the caption of the original images and OCR tokens for the scene-related question. The prompt used for answer paraphrasing can be found in Appendix. Figure 2 illustrates the data modifications we performed on our dataset.

Stage III: Quality Check In this stage, we assign a different annotator to each task to review 10 examples from each split. During this stage, we identify minor format inconsistencies between tasks and address them by standardizing the task formats. We also observe that a few answers (less than 3% of examined instances) were not effectively paraphrased by ChatGPT due to insufficient image information. We employ simple heuristics to filter these paraphrased answers and use a basic template to convert the original answer into a sentence. We find that this small portion of unsuccessful paraphrased answers has negligible impact. Finally, the task dataset is deemed complete once the annotator can successfully load it and re-examine the accuracy of the instructions, inputs, and outputs for each instance examined.

Stage IV: Key Datasets Translation To boost the language diversity and support the evaluation across different languages, we select a subset of datasets (OK-VQA, ImageNet, Winoground, VQAv2, VIST, MSRVTT and MSRVTT-QA) that covers different tasks and translate their evaluation data into 100 languages following FLORES-101 [13]. We translate 500 samples for each split of each task in our first version. More multilingual samples will be supported in the future. We adopt the distillation version NLLB-1.3B [6] for translation, one of the state-of-the-art open multilingual translation models. As there are no native speakers for different languages, we adopt an automatic filtering mechanism to ensure the translation quality, where languages with translation BLEU scores from English larger than 20 based on FLORES-101 results are kept. After this step, only 80 languages are kept (see Appendix for detailed language names).

3.3 Dataset Format

The instance in our dataset consists of five fields: (1) **Images**: we represent the images with the potentially added bounding box by a base64 string. (2) **Instruction**: we randomly select an instruction from the task instruction pool for each instance. (3) **Inputs**: we allocate this field for providing task-specific inputs to the model, e.g., the question in the VQA tasks. For tasks such as captioning, there is no extra input so the corresponding field is left as an empty string. (4) **Outputs**: the required output to the specific tasks, such as the description of the image for captioning tasks and the answer to the image-related question. (5) **Meta Data**: we provide this field to preserve important information such as image id for referencing the original dataset. Figure 3 illustrates an instance in the unified format. With the clear distinction of these fields, the user of our benchmark can flexibly construct the training instances needed and evaluate the models conveniently. Table 3 gives the statistics aggregated by tasks, and we refer readers to Appendix for detailed statistics and the license of each dataset.

List[String]: the base64 string representation of a profile photo of F. Scott Fitzgerald
Images: ["iVBORw0KGg...5ErkJggg=="]

String: task instruction
Instruction: "Analyze the image and provide an appropriate response to the question. "
String: task-specific inputs, e.g., a question related to the image.
Inputs: "On which book by this man, Baz luhrmann's planned a film?"
String: task outputs, e.g., the correct answer for the question.
Outputs: "Baz Luhrmann has planned a film adaptation of the book The Great Gatsby. "
Dict: meta information dictionary contains original data.
Meta Data: {"kilt_id": "qw_1524", ... ,"wikipedia_id": "152171"}

Figure 3: A ViQuAE instance represented in the unified data instance schema used in our dataset.

Table 3: M^{3}IT task descriptions and statistics, encompassing image captioning (CAP), classification (CLS), visual question answering (VQA), knowledgeable visual question answering (KVQA), reasoning (REA), generation (GEN), Chinese vision-language, and video-language tasks. We aggregate instance counts for training, validation, and test sets across all tasks, totaling 2,429,264 instances.

Task	Description	Total #samples		
		Train	Val	Test
CAP	Given an image, write a description for the image.	679,087	41,462	27,499
CLS	Given an image, classify the image into pre-defined categories.	238,303	100,069	21,206
VQA	Given an image, answer a question relevant to the image.	177,633	46,314	10,828
KVQA	Given an image, answer the question requires outside knowledge.	39,981	11,682	5,477
REA	Given an image, conduct reasoning over the images.	99,372	11,500	10,000
GEN	Given an image, make compositions with certain requirements.	145,000	11,315	17,350
Chinese	CAP, CLS, VQA, and GEN tasks in Chinese.	192,076	77,306	4,100
Video	CAP, CLS, and VQA tasks on video-language datasets.	20,868	7,542	9,294
Multi-lingual	Translated tasks in 80 languages	0	240,000	184,000

4 Experiments

In this section, we build a VLM to validate the effectiveness of the proposed M^{3}IT dataset for multi-modal agents. We first introduce the experimental setups (§ 4.1), then report and discuss the results (§ 4.2). Lastly, we analyze the influence of task number and instruction diversity, and provide a qualitative result (§ 4.3).

4.1 Experimental Settings

Implementation Details Inspired by the recent success of BLIP [23], we adopt the vision encoder and the Q-former architecture in the BLIP2-OPT-2.7B [23] model to extract relevant visual features from images. For the large language models, we utilize Ziya-13B [61] derived from LLaMA [49] with bilingual (English and Chinese) ability. We employ a two-staged training. **Stage I Visual-Text Alignment:** To align the visual and textual feature space, we utilize the instructions in the coco captioning and perform an initial alignment training on LAION 400M [41]. We train the Q-former and the language projection, resulting in a total 130M parameters to optimize with AdamW [30]. The batch size is set to 256 to maximize the utilization of GPU and the model is trained with 300k steps. The learning rate linearly increases to a peak value of $5e$-5 in the first 2000 steps and follows a cosine decay scheduler. The weight decay is set to 0.05. **Stage II Multi-modal Instruction Tuning:** We further perform a multi-modal instruction tuning in our benchmark to activate the great potential of LLMs. We train the model after alignment training for 3 epochs and with a lower learning rate of $1e$-5 and a warmup stage of 1000 steps. Inspired by LoRa tuning [16], the weights for mapping query and value vectors in the attention layer of LLMs are learnable in this stage to better adapt to the instruction tuning dataset. Other training parameters are consistent with Stage I. All experiments are conducted with 8 NVIDIA 80GB A100 GPUs. It took about 10 days for Stage I and Stage II can be finished in a day.

Table 4: ROUGE-L evaluation results of KVQA tasks. Our Ying-VLM outperforms all the baselines consistently.

Model	OK-VQA	A-OKVQA	ViQuAE
BLIP2-Flan-T5-XXL	9.1	15.6	9.7
MiniGPT4	23.3	21.8	24.4
InstructBLIP	7.1	5.9	7.3
Ying-VLM (Ours)	**27.5**	**24.5**	**29.6**

Table 5: Zero-shot transfer to Chinese vision-language tasks. Our model generalizes well on unseen Chinese captioning, VQA and classification tasks, with the highest ROUGE-L.

Model	Flickr-8k-CN	FM-IQA	Chinese-FoodNet
MiniGPT4	9.6	20.1	5.0
InstructBLIP	5.2	2.3	1.0
Ying-VLM (Ours)	**20.5**	**33.3**	**49.8**

Table 6: Zero-shot transfer to video-language tasks. We report ROUGE-L score for all tasks.

Model	Video Captioning	Video Question Answer			
	MSRVTT	iVQA	ActivityNet-QA	MSRVTT-QA	MSVD-QA
BLIP-2-Flan-T5-XXL	8.8	11.1	8.9	10.3	13.2
InstructBLIP	**14.3**	6.3	9.3	4.0	7.0
Ying-VLM (Ours)	14.2	**23.5**	**21.9**	**18.3**	**21.4**

Evaluation Setup To examine the generalization of instruction tuning, some tasks are held-out for evaluation (see Figure 1 for held-in/out tasks). We are interested in the following research questions: (RQ1) Can multi-modal instruction tuning elicit world knowledge from LLMs? (RQ2) Can English-only instruction tuning generalize to other languages such as Chinese? and (RQ3) Can image-only multi-modal instruction tuning generalize to video-language tasks? For RQ1, we evaluate our models on three KVQA tasks in our datasets, i.e., OK-VQA [32], A-OKVQA [42] and ViQuAE. For RQ2 and RQ3, we perform zero-shot transfer evaluation on Chinese vision-language and video-language datasets, respectively. We use greedy decoding in inference if not otherwise specified.

Metrics We adopt ROUGE-L [26] as an automatic metric to assess the consistency between predictions and ground-truth answers, focusing on evaluating the model's conversational abilities. As the automatic metric may not fully capture the nuances of conversational quality, we further introduce GPT-4 as a proxy of human evaluators (§ 4.2).

Baselines We compare our models to recently proposed powerful multi-modal agents, including (1) BLIP-2-Flan-T5-XXL [23] where an instruction-tuned Flan-T5 [53] is connected with a powerful vision encoder to perform a series of multi-modal tasks; (2) MiniGPT-4 which aligns a CLIP visual encoder with a frozen Vicuna [5] with artificially collected dialog dataset; and (3) InstructBLIP, a recently proposed instruction tuning enhanced multi-modal agents with Vicuna-13B with converted multi-model datasets and the LLaVA [28] dataset generated by GPT-4.

4.2 Main Results

RQ1: Knowledgeable Visual Question Answer Evaluation The results on the KVQA benchmarks are shown in Table 4. In comparison to the strongest baseline, our model achieves an improvement of 3.2 and 2.7 ROUGE-L points for OK-VQA and A-OKVQA, respectively. Additionally, Ying-VLM delivers the best performance on the held-out ViQuAE dataset. These findings indicate that instruction tuning on M^3IT effectively harnesses knowledge from LLMs and elevates response quality.

RQ2: Zero-Shot Transfer to Chinese Vision-Language Tasks We assess models on three unseen Chinese vision-language tasks to investigate the cross-language generalization capabilities of instruction tuning. BLIP-2 is not considered, as Flan-T5 does not support Chinese.[2] As illustrated in Table 5, our model outperforms MiniGPT4 and InstructBLIP on all evaluated tasks, demonstrating notable improvements. These findings indicate that instruction tuning with English datasets can effectively generalize to different languages, showcasing the promising potential that can be further explored.

RQ3: Zero-Shot Transfer to Video-Language Tasks To evaluate performance on video-language tasks, we uniformly sample 8 frames from each video. A comparison with MiniGPT4 is excluded, as it does not support video inputs. Following the approach of InstructBLIP [7], we concatenate the visual embedding extracted from the Q-former of each frame as a prefix embedding to the language model. As demonstrated in Table 6, our model excels in these challenging settings, significantly

[2]For all models, we introduce a prompt to promote Chinese outputs. See Appendix D for details.

surpassing the BLIP-series baselines. It is worth noting that the training dataset does not include any visual inputs such as videos, implying that our instruction tuning effectively aids the model in generalizing to video inputs with a temporal dimension.

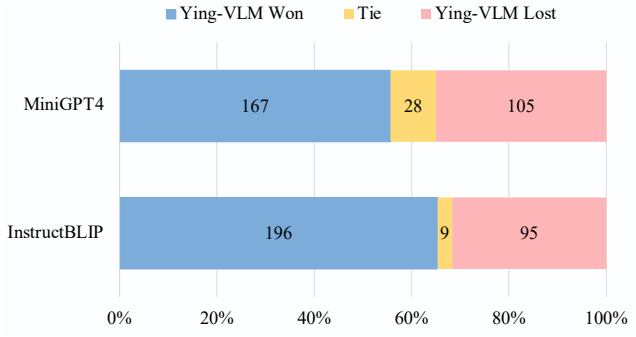

Figure 4: Evaluation results using GPT-4 as an evaluator. Our model outperforms MiniGPT-4 and InstructBLIP with a winning rate at 55.6% and 65.5%, respectively.

GPT-4 Evaluation Results To further validate the quality of the generated response, we propose to utilize the powerful GPT-4 model as a proxy of human evaluators [38, 12]. Specifically, following Vicuna [5], we use GPT-4 to rate the performance of different models against our Ying-VLM. Considering the API cost of GPT-4, 300 examples are randomly sampled from OK-VQA, A-OKVQA and ViQuAE datasets as a subset for evaluation. For each sample, we construct a prompt consisting of the original question, its corresponding reference answer, the response generated by our Ying-VLM, and a baseline system output. GPT-4 is queried with the prompt to rate both responses on a scale of ten based on the given question and its reference answer. The ratings are primarily based on the accuracy, relevance, and naturalness of the response to meet the requirements when humans are interacting with multi-modal agents (see Appendix for the detailed evaluation template). We employ the strategy proposed by Wang et al. [51] to mitigate potential evaluation biases regarding the response order.[3] Figure 4 shows that our Ying-VLM outperforms all baseline models in most samples. Notably, Ying-VLM beat the strongest baseline MiniGPT4 on 167 over 300 tested samples. Consistent with the previous ROUGE-L evaluation, this result indicates that the model fine-tuned on our instruction dataset can produce more accurate and engaging responses on the challenging KVQA tasks.

4.3 Analysis

We investigate the effect of task number and instruction diversity on the performance of learned models, providing insights for future studies to utilize our benchmark better.

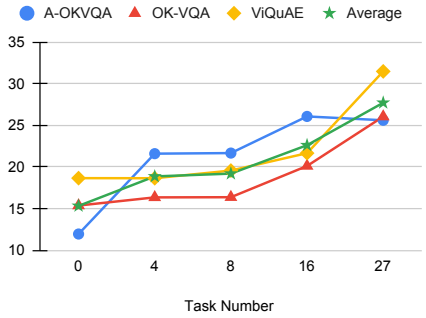

Figure 5: Performance increases with more instruction tuning datasets.

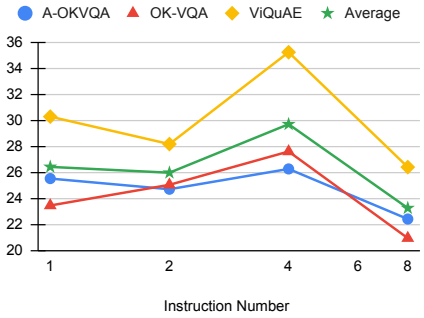

Figure 6: Performance changes with the varied number of instructions used for training.

[3] https://github.com/i-Eval/FairEval

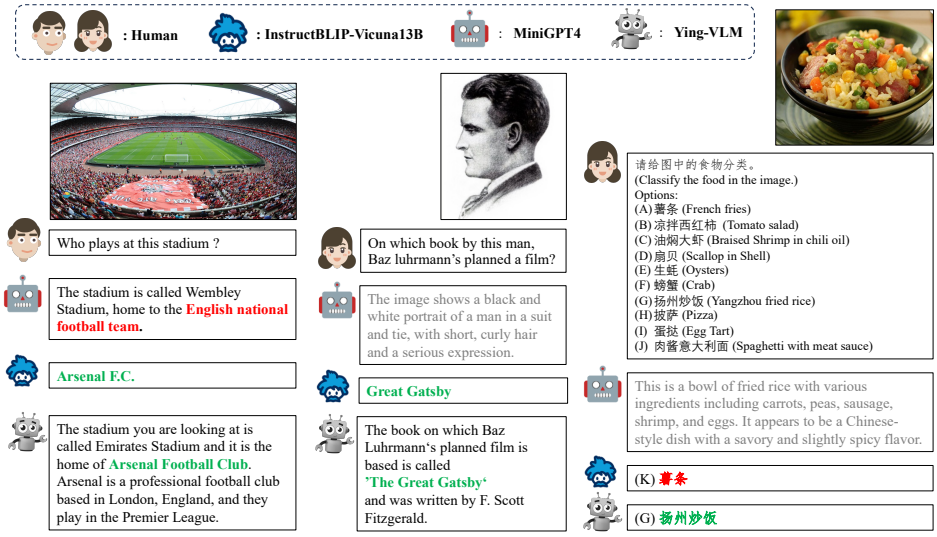

Figure 7: Case study of the model outputs. Correct answers are bolded with green, wrong answers are in red and irrelevant answers are in grey. The model trained with our datasets can provide natural and informative responses to entity-centric questions, and generalize to the food classification task in Chinese (English translation for visualization only).

Effect of Task Number We investigate the influence of task numbers by randomly shuffling our tasks and then selecting a subset to train the model during the instruction tuning stage. Due to the computational resource limitation, we set up a maximum of 5k examples for each task and train all the models for 5k steps with a batch size of 64. We select 0, 4, 8, 16 and all 27 tasks for training, and report the individual ROUGE-L score and the average score. As illustrated in Figure 5, increasing the number of tasks greatly improves the results of the generalization performance. Besides, the performance gain is not diminished as the task number increases. This is promising as it indicates that we can continually improve performance by introducing more tasks into the training. It would be interesting to investigate the influence of different task clusters, which we leave for future studies.

Effect Instruction Diversity To investigate the influence of instruction diversity, we limit the number of instructions used in each dataset to 1, 2, 4, and 8, resulting in varying levels of diversity for each task. The other training parameters are consistent with those used in previous experiments on task number investigation. Figure 6 shows that the performance varies with the level of diversity. Specifically, our results suggest that using four instructions per task is sufficient for achieving decent performance. We leave a more in-depth analysis of the instruction diversity for future work.

Qualitative Results We conduct a case study to provide a more straightforward understanding of instruction-tuned models. The cases are chosen from the held-out ViQuAE and ChineseFoodNet datasets. As shown in Figure 7, our model generates accurate responses to all questions. In contrast, MiniGPT4 produces an incorrect answer for the stadium question on the left and fails to follow instructions in the subsequent cases, providing generic image descriptions instead. Additionally, compared to InstructBLIP, which provides concise but less engaging answers for the two questions requiring external knowledge, our model responds more naturally and engagingly, underlining the value of our dataset. Our model also successfully generalizes to Chinese inputs, accurately classifying the food image based on the instruction. These cases emphasize the importance of instruction tuning and demonstrate that our dataset can effectively enhance the capabilities of VLMs.

5 Conclusion

In this paper, we present M^3IT, a multi-modal multilingual instruction tuning dataset for aiding the development of multi-modal large language models. The dataset comprises 2.4 million carefully curated instances and 400 manually written task instructions across 40 tasks. We build Ying-VLM to validate the effectiveness of our dataset. Quantitative and qualitative results demonstrate that the

models trained with our datasets successfully follow human instructions, provide more engaging responses, and achieve strong generalization performance on unseen video and Chinese tasks. Further analysis shows that the increased task number can continually boost performance, and instruction diversity can influence results. We hope our proposed benchmark, trained models, and experimental findings can facilitate future studies toward building powerful multi-modal intelligent agents.

A Dataset Statistics

Table 7: Detailed task descriptions and statistics of our instruction tuning tasks, including all datasets in all types of tasks. The column "Used" indicates whether we use this dataset in the instruction tuning stage.

Task	Dataset	Used	#samples			License
			Train	Val	Test	
Captioning	MS COCO [27]	Yes	566,747	25,010	25,010	Custom
	TextCaps [44]	Yes	97,765	13,965	0	Unknown
	Image-Paragraph-Captioning [21]	Yes	14,575	2,487	2,489	Custom
Classification	COCO-GOI [27]	Yes	30,000	2,000	0	Custom
	COCO-Text [50]	Yes	118,312	27,550	0	Custom
	ImageNet [40]	Yes	30,000	50,000	0	Non-commercial
	COCO-ITM [27]	Yes	30,000	5,000	5,000	Custom
	e-SNLI-VE [20]	Yes	20,000	14,339	14,740	Unknown
	Mocheg [58]	Yes	4,991	180	466	CC BY 4.0
	IQA [9]	Yes	5,000	1,000	1,000	Custom
VQA	VQA v2 [15]	Yes	30,000	30,000	0	CC-BY 4.0
	Shapes VQA [1]	Yes	13,568	1,024	1,024	Unknown
	DocVQA [33]	Yes	39,463	5,349	0	Unknown
	OCR-VQA [34]	Yes	11,414	4,940	0	Unknown
	ST-VQA [2]	Yes	26,074	0	4,070	Unknown
	Text-VQA [45]	Yes	27,113	0	5,734	CC BY 4.0
	GQA [18]	Yes	30,001	5,001	0	CC BY 4.0
KVQA	OK-VQA [32]	Yes	9,009	5,046	0	Unknown
	A-OK-VQA [42]	Yes	17,056	1,145	0	Unknown
	ScienceQA [31]	Yes	12,726	4,241	4,241	CC BY-NC-SA
	ViQuAE [22]	No	1,190	1,250	1,236	CC By 4.0
Reasoning	CLEVR [19]	Yes	30,000	2,000	0	CC BY 4.0
	NLVR [46]	Yes	29,372	2,000	0	Unknown
	VCR [60]	Yes	25,000	5,000	5,000	Custom
	VisualMRC [47]	Yes	15,000	2,500	5,000	Unknown
	Winoground [48]	No	0	0	800	Unknown
Generation	Visual Storytelling [17]	Yes	5,000	4,315	4,350	Unknown
	Visual Dialog [8]	Yes	50,000	1,000	1,000	CC By 4.0
	Multi30k [10]	Yes	90,000	6,000	12,000	Non-commercial
Chinese	FM-IQA [11]	No	164,735	75,206	0	Unknown
	COCO-Caption CN [25]	No	18,341	1,000	1,000	Non-commercial
	Flickr-8k-Caption CN [24]	No	6,000	1,000	1,000	CC By 3.0
	Chinese Food Classification [4]	No	0	0	1,100	Unknown
	Multimodal Chat [62]	No	3,000	1,000	1,000	Unknown
Video	Action-Classification [14]	No	2,000	2,000	2,000	Custom
	iVQA [57]	No	5,994	2,000	2,000	Unknown
	MSVD QA [54]	No	1,161	245	504	Unknown
	ActivityNet QA [59]	No	3,200	1,800	800	Unknown
	MSRVTT QA [54]	No	6,513	497	2,990	Unknown
	MSRVTT Captioning [55]	No	2,000	1,000	1,000	Unknown

Table 7 lists the detailed statistics in our benchmark. We collect the dataset license from PaperWithCode.[4] For datasets under Unknown and Custom licenses, we suggest the users check the project page or contact the dataset owner before usage.

[4] `https://paperswithcode.com/`

B Template for Answer Paraphrase

We provide the paraphrase template in Table 8 for querying the ChatGPT to re-write the original short answers, where {Q} and {A} is filled with the question and the answer need to be paraphrased, respectively. We incorporate an example to better inform the model of the paraphrasing tasks. For VQAv2 tasks, we add an extra {Caption} field in the template filled with corresponding captions from the COCO dataset to provide extra context information to help to paraphrase.

Table 8: Template used to query ChatGPT for answer paraphrasing.

> You are an AI visual assistant. Now you are given a question related to an image and a short ground-truth answer. Your task is to transform the ground-truth answer into a natural and convincing response. Make sure the response is accurate, highly relevant to the question, and consistent with the original answer.
>
> Question:
> Which NASA space probe was launched to this planet in 1989?
> Answer:
> Magellan
> Transformed Answer:
> NASA sent the Magellan spacecraft to Venus in 1989, which was the first planetary spacecraft launched from a space shuttle.
>
> Question:
> {Q}
> Answer:
> {A}
> Transformed Answer:

C Dataset Translation

We translate all the task instructions and evaluation sets of ImageNet, Winoground, VQAv2, OK-VQA, VIST, MSRVTT and MSRVTT-QA into 80 languages, as shown in Table 9. Due to the computational resource constraint, we translate the whole test of Winoground (800 examples) and set a maximum instance number of 500 for each split in other tasks.

D Prompt for Zero-Shot Chinese Vision-Language Tasks

In our experiments, all Vision-Language models are fine-tuned exclusively using English data. In our preliminary study, we observe that these models tend to generate English responses, even when the input and instructions are written in Chinese. We introduce a simple Chinese dialogue context during the zero-shot Chinese Vision-Language Task evaluation for all models, as illustrated in Table 10, Interestingly, this minor adjustment can encourage models to produce reasonable Chinese output. We leave the analysis of instruction-tuned VLM models' multilingual capabilities for future research.

E Template for GPT-4 Evaluation

We adopt the template in Table 11 to query GPT-4 and obtain the evaluation results with FairEval [5] to obtain more stable results. Specifically, each tested instance is a quaternion: `(question, reference, response1, response2)`, where `response1` and `response2` are two responses from our Ying-VLM and the baseline model, respectively. For each instance, we query GPT-4 to judge which response is of better quality regarding accuracy, relevance and naturalness. We populate the quaternion into the evaluation template to form

[5] `https://github.com/i-Eval/FairEval`

Table 9: List of Language Codes, Scripts, and Languages Names for translated datasets.

Language Code	Script	Language Name
af	afr_Latn	Afrikaans
am	amh_Ethi	Amharic
ar	arb_Arab	Modern Standard Arabic
as	asm_Beng	Assamese
ast	ast_Latn	Asturian
be	bel_Cyrl	Belarusian
bg	bul_Cyrl	Bulgarian
bn	ben_Beng	Bengali
bs	bos_Latn	Bosnian
ca	cat_Latn	Catalan
ceb	ceb_Latn	Cebuano
cs	ces_Latn	Czech
cy	cym_Latn	Welsh
da	dan_Latn	Danish
de	deu_Latn	German
el	ell_Grek	Greek
es	spa_Latn	Spanish
et	est_Latn	Estonian
fi	fin_Latn	Finnish
fr	fra_Latn	French
fuv	fuv_Latn	Nigerian Fulfulde
gl	glg_Latn	Galician
gu	guj_Gujr	Gujarati
ha	hau_Latn	Hausa
he	heb_Hebr	Hebrew
hi	hin_Deva	Hindi
hr	hrv_Latn	Croatian
hu	hun_Latn	Hungarian
hy	hye_Armn	Armenian
id	ind_Latn	Indonesian
ig	ibo_Latn	Igbo
is	isl_Latn	Icelandic
it	ita_Latn	Italian
ja	jpn_Jpan	Japanese
jv	jav_Latn	Javanese
ka	kat_Geor	Georgian
kk	kaz_Cyrl	Kazakh
km	khm_Khmr	Khmer
kn	kan_Knda	Kannada
ko	kor_Hang	Korean
ky	kir_Cyrl	Kyrgyz
lb	ltz_Latn	Luxembourgish
lg	lug_Latn	Ganda
lij	lij_Latn	Ligurian
li	lim_Latn	Limburgish
ln	lin_Latn	Lingala
lo	lao_Laoo	Lao
lt	lit_Latn	Lithuanian
lv	lvs_Latn	Standard Latvian
mi	mri_Latn	Maori
mk	mkd_Cyrl	Macedonian
ml	mal_Mlym	Malayalam
mr	mar_Deva	Marathi
mt	mlt_Latn	Maltese
my	mya_Mymr	Burmese
nl	nld_Latn	Dutch
ny	nya_Latn	Nyanja
oc	oci_Latn	Occitan
pa	pan_Guru	Eastern Panjabi
pl	pol_Latn	Polish
pt	por_Latn	Portuguese
ro	ron_Latn	Romanian
ru	rus_Cyrl	Russian
sd	snd_Arab	Sindhi
sk	slk_Latn	Slovak
sn	sna_Latn	Shona
so	som_Latn	Somali
sr	srp_Cyrl	Serbian
sv	swe_Latn	Swedish
ta	tam_Taml	Tamil
te	tel_Telu	Telugu
tg	tgk_Cyrl	Tajik
th	tha_Thai	Thai
tl	tgl_Latn	Tagalog
tr	tur_Latn	Turkish
uk	ukr_Cyrl	Ukrainian
ur	urd_Arab	Urdu
vi	vie_Latn	Vietnamese
wo	wol_Latn	Wolof
zh	zho_Hans	Chinese (Simplified)

two query prompts: `T(Q=question, R=reference, R1=response1, R2=response2)` and `T(Q=question, R=reference, R1=response2, R2=response1)`. We set the temperature of GPT-4 to 1 and sample three completions for each query prompt. Therefore, each response will

Table 10: Prompt for promoting Chinese outputs.

<human>: 请根据我的指示，以及所给的图片，做出相应的回答。
<bot>:
好的。
<human>:
{Instruction}
{Input}
<bot>:
好的。

receive 6 scores, and we use the average score as the final score for each response. The response with the higher final score is considered the better response. The GPT-4 evaluation incurred a cost of $20.45 for InstructBlip and $20.90 for MiniGPT-4.

Table 11: Template used to query GPT-4 for evaluating the response quality of different models.

[Question]
{Q}
[The Start of Reference Answer]
{R}
[The End of Reference Answer]
[The Start of Assistant 1's Answer]
{R1}
[The End of Assistant 1's Answer]
[The Start of Assistant 2's Answer]
{R2}
[The End of Assistant 2's Answer]

[System]
We would like to request your feedback on the performance of two AI assistants in response to the user's multimodal question displayed above. We provided no multimodal inputs other than question text, but we provided a reference answer for this question. You need to evaluate the quality of the two responses based on the question and the reference answer.
Please rate the on the follow aspects:
1. Accuracy: whether the candidate's response is consistent with the original answer, this is important as we do not want a misleading result;
2. Relevance: whether the candidate's response is highly relevant to the question and image content;
3. Naturalness: whether the candidate's response is engaging, providing a great communication experience for the user when interacting with the AI visual assistant.
of the two Assistants' responses.

Each assistant receives an overall score on a scale of 1 to 10, where a higher score indicates better overall performance.
Please first provide a comprehensive explanation of your evaluation, avoiding any potential bias and ensuring that the order in which the responses were presented does not affect your judgment.
Then, output two lines indicating the scores for Assistant 1 and 2, respectively.

Output with the following format:
Evaluation evidence: <evaluation explanation here>
The score of Assistant 1: <score>
The score of Assistant 2: <score>

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